Supplementary Information of "Cavity-enhanced solid-state nuclear spin gyroscope"

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I. NUCLEAR SPIN STATE MIXING

A single NV center in the ground state comprises an $S = 1$ electronic triplet state coupled to the $I = 1$ ^{14}N nuclear states. Taking the NV direction as the quantization axis, the NV center Hamiltonian reads:

$$H_{\mathrm{NV}}/h = DS_z^2 + \gamma_e \hat{\mathbf{S}} \cdot \mathbf{B} + Q\hat{I}_z^2 - \gamma_n \hat{\mathbf{I}} \cdot \mathbf{B} + \hat{\mathbf{S}} \cdot A \cdot \hat{\mathbf{I}} \tag{1}$$

where $D = 2.87$ GHz and $Q = -4.945$ MHz are the electron and nuclear spin zero-field splittings, $\gamma_e = 2.802$ MHz/G and $\gamma_n = 0.308$ kHz/G are the electron and nuclear spin gyromagnetic factors, respectively, and A is the diagonal hyperfine interaction tensor with $A_{zz} = -2.162$ MHz and $A_{xx} = A_{yy} = -2.62$ MHz. Rewrite it with creation and

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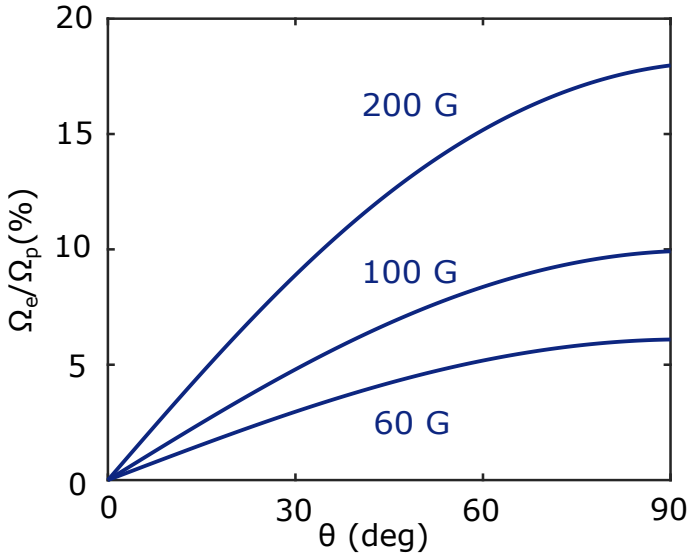


Supplementary Figure 1. Rabi frequency ratio for the spin-exchanging transition (Ω_e) to a spin-preserving transition (Ω_p) with the off-axis magnetic field-induced state mixing.

annihilation operators:

$$\begin{aligned} H_{\mathrm{NV}}/h = & DS_z^2 + \gamma_e B_z S_z + \gamma_e (B_+ S_+ + B_- S_-) + QI_z^2 - \gamma_n B_z I_z \\ & - \gamma_n (B_+ I_+ + B_- I_-) + \frac{1}{2} A_{xx} (S_+ I_+ + S_+ I_- + S_- I_+ + S_- I_-) \\ & - \frac{1}{2} A_{yy} (S_+ I_+ - S_+ I_- - S_- I_+ + S_- I_-) + A_{zz} S_z I_z \end{aligned} \tag{2}$$

Considering $A_{xx} = A_{yy} = A_\perp$ We have:

$$\begin{aligned} H_{\mathrm{NV}}/h = & DS_z^2 + \gamma_e B_z S_z + \gamma_e (B_+ S_+ + B_- S_-) + QI_z^2 - \gamma_n B_z I_z \\ & - \gamma_n (B_+ I_+ + B_- I_-) + A_\perp (S_+ I_- + S_- I_+) + A_{zz} S_z I_z \end{aligned} \tag{3}$$

Applying the Hamiltonian to the eigenstates:

$$\begin{aligned} H_{\mathrm{NV}}|0_e, 0_I\rangle &= \gamma_e B_+ |1_e, 0_I\rangle - \gamma_n B_+ |0_e, 1_I\rangle - \gamma_n B_- |0_e, -1_I\rangle + A_\perp |1_e, -1_I\rangle + \gamma_e B_- |1_e, 0_I\rangle \\ H_{\mathrm{NV}}|0_e, 1_I\rangle &= \gamma_e B_+ |1_e, 1_I\rangle + (Q - \gamma_n B_z)|0_e, 1_I\rangle - \gamma_n B_- |0_e, 0_I\rangle + A_\perp |1_e, 0_I\rangle \\ H_{\mathrm{NV}}|0_e, -1_I\rangle &= \gamma_e B_+ |1_e, -1_I\rangle + (Q + \gamma_n B_z)|0_e, -1_I\rangle - \gamma_n B_+ |0_e, 0_I\rangle \\ H_{\mathrm{NV}}|1_e, 0_I\rangle &= (D + \gamma_e B_z)|1_e, 0_I\rangle + \gamma_e B_- |0_e, 0_I\rangle - \gamma_n B_+ |1_e, 1_I\rangle - \gamma_n B_- |1_e, -1_I\rangle + A_\perp |0_e, 1_I\rangle \\ H_{\mathrm{NV}}|1_e, 1_I\rangle &= (D + \gamma_e B_z + Q - \gamma_n B_z + A_{zz})|1_e, 1_I\rangle + \gamma_e B_- |0_e, 1_I\rangle - \gamma_n B_- |1_e, 0_I\rangle \\ H_{\mathrm{NV}}|1_e, -1_I\rangle &= (D + \gamma_e B_z + Q + \gamma_n B_z - A_{zz})|1_e, -1_I\rangle + \gamma_e B_- |0_e, -1_I\rangle - \gamma_n B_+ |1_e, 0_I\rangle + A_\perp |0_e, 0_I\rangle \end{aligned} \tag{4}$$

Now we diagonalize the matrix to see the overlap between new wavefunctions. The ratio between the nuclear spin exchanging transition rate Ω_e and the nuclear spin preserving transition rate Ω_p is shown in Fig. S1. The state mixing enables the second field driving and the electromagnetically induced transparency (EIT) and masing without inversion (MWI) discussions.

II. TWO FIELD INTERFERENCE IN A HYBRID QUANTUM SYSTEM

A. EIT within a Λ system with optical polarization

In this section, we introduce the EIT within a Λ system (Fig. S2(a) inset) with optical polarization. We consider a homogeneous spin ensemble. Since there is no interaction with a cavity mode, all the driving of the spins is from a fixed and effective classical field. Because of this, each NV center in the ensemble behaves independently, and thus, as is usual in EIT analyses, we can simply treat the evolution of a single NV. The resulting signal is simply the sum of the contribution from all the identical NV centers. The Hamiltonian for a single NV center reads [1]:

$$\mathcal{H} = \omega_2\sigma_{22} + \omega_e\sigma_{ee} + \Omega_1\left(\sigma_{1e}e^{-i(\omega_e+\Delta_1)t} + \sigma_{e1}e^{i(\omega_e+\Delta_1)t}\right) + \Omega_2\left(\sigma_{2e}e^{-i(\omega_e-\omega_2+\Delta_2)t} + \sigma_{e2}e^{i(\omega_e-\omega_2+\Delta_2)t}\right) \tag{5}$$

with ω_2 the energy of the state $|2\rangle$, ω_e the frequency of the state $|e\rangle$, Ω_1 (Ω_2) is the Rabi frequency for the driving field 1 (2), respectively. Δ_1 (Δ_2) is the detuning of driving field 1 (2). The time-dependence can be removed by transforming the wavefunction according to $|\psi\rangle \to \exp(-i(\omega_e + \Delta_1)t\sigma_{ee})|\psi\rangle$:

$$\mathcal{H} = \omega_2\sigma_{22} - \Delta_1\sigma_{ee} + \Omega_1\left(\sigma_{1e} + \sigma_{e1}\right) + \Omega_2\left(\sigma_{2e}e^{-i(-\omega_2+\Delta_2-\Delta_1)t} + \sigma_{e2}e^{i(-\omega_2+\Delta_2-\Delta_1)t}\right) \tag{6}$$

Then by: $|\psi\rangle \to \exp(-i(-\omega_2 + \Delta_2 - \Delta_1)t\sigma_{22})|\psi\rangle$:

$$\mathcal{H} = (\Delta_2 - \Delta_1)\sigma_{22} - \Delta_1\sigma_{ee} + \Omega_1\left(\sigma_{1e} + \sigma_{e1}\right) + \Omega_2\left(\sigma_{2e} + \sigma_{e2}\right) \tag{7}$$

We assume the ensemble is homogeneous for simpleness:

$$\mathcal{H} = (\Delta_2 - \Delta_1)\sigma_{22} - \Delta_1\sigma_{ee} + \Omega_1\left(\sigma_{1e} + \sigma_{e1}\right) + \Omega_2\left(\sigma_{2e} + \sigma_{e2}\right) \tag{8}$$

The Lindbladian for the system contains the dephasing rate of the electron spin transition Γ, the spin polarization rate γ_p, and thermal relaxation rate γ_{th}. Combining the Hamiltonian and Lindbladian operators, the Maxwell-Bloch equation set reads:

$$\begin{aligned}
\frac{d\sigma_{1e}}{dt} &= i\Delta_1\sigma_{1e} - i\Omega_1(\sigma_{11} - \sigma_{ee}) - i\Omega_2\sigma_{12} - \frac{\Gamma}{2}\sigma_{1e} \\
\frac{d\sigma_{2e}}{dt} &= i\Delta_2\sigma_{2e} - i\Omega_1\sigma_{21} - i\Omega_2(\sigma_{22} - \sigma_{ee}) - \frac{\Gamma}{2}\sigma_{2e} \\
\frac{d\sigma_{12}}{dt} &= -i(\Delta_2 - \Delta_1)\sigma_{12} + i\Omega_1\sigma_{e2} - i\Omega_2\sigma_{1e} - \gamma_{\text{th}}\sigma_{12} \\
\frac{d\sigma_{11}}{dt} &= i\Omega_1(\sigma_{e1} - \sigma_{1e}) + \gamma_{\text{th}}(\sigma_{ee} + \sigma_{22} + \sigma_{33} - 3\sigma_{11}) + \gamma_p\sigma_{ee} \\
\frac{d\sigma_{22}}{dt} &= -i\Omega_2(\sigma_{2e} - \sigma_{e2}) + \gamma_{\text{th}}(\sigma_{ee} + \sigma_{11} + \sigma_{33} - 3\sigma_{22}) + \gamma_p\sigma_{33} \\
\frac{d\sigma_{33}}{dt} &= \gamma_{\text{th}}(\sigma_{11} + \sigma_{22} + \sigma_{ee} - 3\sigma_{33}) - \gamma_p\sigma_{33}
\end{aligned} \tag{9}$$

From the subequations (9) and $\sigma_{11}+\sigma_{22}+\sigma_{33}+\sigma_{ee}=1$, we can simplify the results with a limit $\Gamma \gg \gamma_{\rm p} \gg \gamma_{\rm th}$:

$$\begin{aligned}
A &= -i\Omega_1(\sigma_{1e}-\sigma_{e1}) \\
B &= -i\Omega_2(\sigma_{2e}-\sigma_{e2}) \\
\sigma_{11} &= -\frac{\gamma_{\rm p}B-4\gamma_{\rm th}A-2\gamma_{\rm p}\gamma_{\rm th}-4\gamma_{\rm th}^2}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})} \\
\sigma_{22} &= -\frac{-(\gamma_{\rm p}+4\gamma_{\rm th})B-2\gamma_{\rm p}\gamma_{\rm th}-4\gamma_{\rm th}^2}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})} \\
\sigma_{ee} &= -\frac{A+B-\gamma_{\rm th}}{\gamma_{\rm p}+4\gamma_{\rm th}} \\
\sigma_{11}-\sigma_{ee} &= \frac{8\gamma_{\rm th}A+(-\gamma_{\rm p}+4\gamma_{\rm th})B+2\gamma_{\rm p}\gamma_{\rm th}}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})} \\
&\approx \frac{2A}{\gamma_{\rm p}}-\frac{B}{4\gamma_{\rm th}}+\frac{1}{2} \\
\sigma_{22}-\sigma_{ee} &= \frac{4\gamma_{\rm th}A+(\gamma_{\rm p}+8\gamma_{\rm th})B+2\gamma_{\rm p}\gamma_{\rm th}}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})} \\
&\approx \frac{A}{\gamma_{\rm p}}+\frac{B}{4\gamma_{\rm th}}+\frac{1}{2}
\end{aligned} \tag{10}$$

The equation simplification finally gives:

$$\begin{aligned}
i\Delta_2\Omega_2(\sigma_{2e}-\sigma_{e2})+i\Delta_1\Omega_1(\sigma_{1e}-\sigma_{e1}) &= \frac{\Gamma}{2}[\Omega_1(\sigma_{1e}+\sigma_{e1})+\Omega_2(\sigma_{2e}+\sigma_{e2})] \\
-\Delta_2 B-\Delta_1 A &= \frac{\Gamma}{2}[\Omega_1(\sigma_{1e}+\sigma_{e1})+\Omega_2(\sigma_{2e}+\sigma_{e2})]
\end{aligned} \tag{11}$$

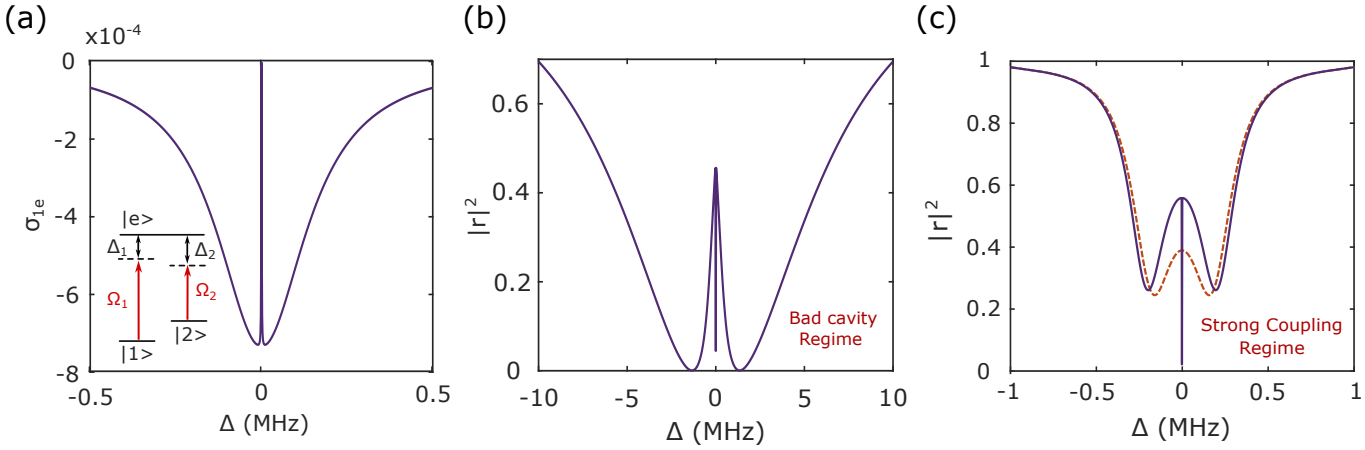
We have one real number σ_{11}, two complex variables σ_{1e} and σ_{2e}, assuming:

$$\begin{aligned}
\sigma_{1e} &= \alpha_1+i\beta_1 \\
\sigma_{2e} &= \alpha_2+i\beta_2
\end{aligned} \tag{12}$$

We have equations below:

$$\begin{aligned}
A &= 2\Omega_1\beta_1 \\
B &= 2\Omega_2\beta_2 \\
\sigma_{12} &= \frac{i[\Omega_1(\alpha_2-i\beta_2)-\Omega_2(\alpha_1+i\beta_1)]}{i(\Delta_2-\Delta_1)+\gamma_{\rm th}} \\
0 &= i\Delta_1(\alpha_1+i\beta_1)-i\Omega_1[\frac{8\gamma_{\rm th}A+(-\gamma_{\rm p}+4\gamma_{\rm th})B+2\gamma_{\rm p}\gamma_{\rm th}}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})}]-i\Omega_2\sigma_{12}-\frac{\Gamma}{2}(\alpha_1+i\beta_1) \\
0 &= i\Delta_2(\alpha_2+i\beta_2)-i\Omega_1\sigma_{21}-i\Omega_2[\frac{4\gamma_{\rm th}A+(\gamma_{\rm p}+8\gamma_{\rm th})B+2\gamma_{\rm p}\gamma_{\rm th}}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})}]-\frac{\Gamma}{2}(\alpha_2+i\beta_2) \\
-\Delta_2 B-\Delta_1 A &= \frac{\Gamma}{2}[\Omega_1(2\alpha_1)+\Omega_2(2\alpha_2)] \\
\sigma_{11} &= -\frac{\gamma_{\rm p}B-4\gamma_{\rm th}A-2\gamma_{\rm p}\gamma_{\rm th}-4\gamma_{\rm th}^2}{4\gamma_{\rm th}(\gamma_{\rm p}+4\gamma_{\rm th})}
\end{aligned} \tag{13}$$

We plot the imaginary part of the coherence σ_{1e} with parameters $\Gamma = 2\pi\times 320$ kHz, $\gamma_{\rm th} = 2\pi\times 80$ Hz, $\gamma_p = 2\pi\times 1$ kHz, $\Omega_1 = 2\pi\times 10$ Hz, $\Omega_2 = 2\pi\times 2$ kHz in Fig. S2(a). A transparency window shows up around $\Delta = 0$ with the linewidth as nuclear spin coherence. We note that even without cavity amplification, in a three-level system, it is already possible to see the positive $\mathrm{Im}(\sigma_{1e})$, or equivalently, the MWI [2]. However, the MWI effect is tiny without the cavity amplification.



Supplementary Figure 2. **EIT in different setups.** (a) EIT without cavity. Two classical fields, Ω_1 and Ω_2, are applied on the three-level system. A transparency window shows up around $\Delta = 0$. (b) EIT in the bad cavity regime with parameters: $\kappa = 2\pi \times 2.6$ MHz, $\kappa_{c1} = 2\pi \times 1.3$ MHz, $g_s = 2\pi \times 0.06$ Hz, $N = 2 \times 10^{14}$, $\Gamma = 2\pi \times 330$ kHz, $\gamma_{\rm th} = 2\pi\times$ 0.08 kHz. A DIR shows up with the electron spin linewidth Γ around $\Delta = 0$ with a transparency window with a nuclear spin linewidth $\gamma_{nd} = 2\pi \times 0.08$ kHz. (c) EIT in the strong coupling regime with parameters: $\kappa = 2\pi \times 0.26$ MHz, $\kappa_{c1} = 2\pi \times 0.13$ MHz, $g_s = 2\pi \times 0.019$ Hz, $N = 2 \times 10^{14}$, $\Gamma = 2\pi \times 330$ kHz. Instead of DIR, an avoid crossing feature shows up with polariton splitting $2g$. Dashed line: Cavity reflection $|r|^2$ without the driving field.

B. EIT and MWI within a cavity coupled system

In this section, we introduce the EIT and MWI in a cavity-coupled hybrid quantum system. We start from the three-level system Hamiltonian:

$$\mathcal{H} = \omega_c a^\dagger a + \sum_j \omega_e^{(j)} \sigma_{ee}^{(j)} + \sum_j g_s^{(j)} \left[a\sigma_{e1}^{(j)} + a^\dagger \sigma_{1e}^{(j)} \right] + \sum_j \Omega_2 \left(\sigma_{2e}^{(j)} e^{i\omega_2 t} + \sigma_{e2}^{(j)} e^{-i\omega_2 t} \right) + \sum_j Q^{(j)} \sigma_{22}^{(j)} + iJ \left(a^\dagger e^{-i\omega_d t} - a e^{i\omega_d t} \right) \tag{14}$$

with ω_c the cavity frequency, $\omega_e^{(j)}$ is the transition frequency between the $|1\rangle$ and $|e\rangle$ for j-th spin, $g_s^{(j)}$ is the single spin coupling strength, Ω_2 the driving field Rabi frequency, ω_2 is the frequency of the driving field, ω_d is the probe frequency, $Q^{(j)}$ is frequency difference between $|2\rangle$ and $|1\rangle$ for j-th spin, and J is the probe strength ($J = \sqrt{\kappa_{c1} P/\hbar\omega_d} = \sqrt{\kappa_{c1} V^2/\hbar\omega_d R}$, with R the resistance, and V the voltage input to the system). The time-dependence can be removed by transforming the wavefunction according to $|\psi\rangle \to \exp\Big(-i\omega_d t(a^\dagger a + \sum_j \sigma_{ee}^{(j)})\Big)|\psi\rangle$:

$$\mathcal{H} = \Delta a^\dagger a + \sum_j \Delta_s^{(j)} \sigma_{ee}^{(j)} + \sum_j g_s^{(j)} \left[a\sigma_{e1}^{(j)} + a^\dagger \sigma_{1e}^{(j)} \right] + \sum_j \Omega_2 \left(\sigma_{2e}^{(j)} e^{i(\omega_2-\omega_d)t} + \sigma_{e2}^{(j)} e^{-i(\omega_2-\omega_d)t} \right) + \sum_j Q^{(j)} \sigma_{22}^{(j)} + iJ \left(a^\dagger - a \right) \tag{15}$$

with $\Delta = \omega_c - \omega_d$ the cavity-probe detuning, $\Delta_s = \omega_s - \omega_d$ the spin-probe detuning. Then by another unitary transformation: $|\psi\rangle \to \exp(i(\omega_2 - \omega_d)t\sigma_{22})|\psi\rangle$:

$$\mathcal{H} = \Delta a^\dagger a + \sum_j \Delta_s^{(j)} \sigma_{ee}^{(j)} + \sum_j \Delta_2^{(j)} \sigma_{22}^{(j)} + \sum_j g_s^{(j)} \left[a\sigma_{e1}^{(j)} + a^\dagger \sigma_{1e}^{(j)} \right] + \sum_j \Omega_2 \left(\sigma_{2e}^{(j)} + \sigma_{e2}^{(j)} \right) + iJ \left(a^\dagger - a \right) \tag{16}$$

with $\Delta_2 = \omega_2 + |Q^{(j)}| - \omega_d$. Before transferring to the Maxwell-Bloch equations, we first apply some approximations. (1) We assume the single spin coupling strength $g_s^{(j)}$ is the same for all spins, i.e., $g_s^{(j)} \to g_s$. It is a valid treatment

as long as the cavity field is homogeneous for the spin ensemble. This is demonstrated in the previous NV-cQED experiments and cavity design [3]. (2). We neglect the nuclear spin inhomogeneity, i.e., $\sum_j Q^{(j)}\sigma_{22}^{(j)} \to Q\sigma_{22}$. Inhomogeneity across the diamond sample is largely attributed to local differences in the magnetic environment due to the ^{13}C nuclear spin bath, which can cause significant shifts in electron spin energies (on the order of 100 kHz). The nuclear spin gyromagnetic ratio is reduced by a factor of 10^4, resulting in expected magnetic-field inhomogeneity on the order of 10 Hz. This is less than the observed effect of electron-spin T_1 processes, which cause random dephasing events at a rate of $\gamma_{\rm th} \sim 100$ Hz. We approximate the nuclear spin coherence as a pure dephasing process as a consequence, though EIT effects could potentially be used to investigate the coherence properties of NV nuclear spins. After these two treatments, the Hamiltonian reads:

$$\mathcal{H} = \Delta a^\dagger a + \sum_j \Delta_s^{(j)}\sigma_{ee}^{(j)} + \Delta_2\sigma_{22} + \sum_j g_s\left[a\sigma_{e1}^{(j)} + a^\dagger\sigma_{1e}^{(j)}\right] + \sum_j \Omega_2\left(\sigma_{2e}^{(j)} + \sigma_{e2}^{(j)}\right) + iJ\left(a^\dagger - a\right) \tag{17}$$

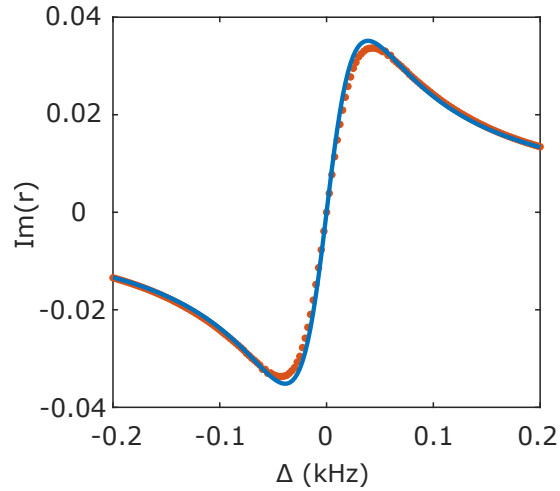
The Maxwell-Bloch equations of motions for the cavity field, spin coherence, and polarization are:

$$\begin{aligned}
\frac{da}{dt} &= -i\Delta a - i\sum_j g_s\sigma_{1e}^{(j)} + J - \frac{\kappa}{2}a \\
\frac{d\sigma_{1e}^{(j)}}{dt} &= -i\Delta_s^{(j)}\sigma_{1e}^{(j)} - ig_s a(\sigma_{11} - \sigma_{ee}^{(j)}) - i\Omega_2\sigma_{12} - \frac{\Gamma}{2}\sigma_{1e}^{(j)} \\
\frac{d\sigma_{2e}^{(j)}}{dt} &= -i(\Delta_s^{(j)} - \Delta_2)\sigma_{2e}^{(j)} - ig_s a\sigma_{21} - i\Omega_2(\sigma_{22} - \sigma_{ee}^{(j)}) - \frac{\Gamma}{2}\sigma_{2e}^{(j)} \\
\frac{d\sigma_{12}}{dt} &= -i\Delta_2\sigma_{12} + ig_s a\sigma_{e2}^{(j)} - i\Omega_2\sigma_{1e}^{(j)} - \frac{\gamma_{\rm nd}}{2}\sigma_{12} \\
\frac{d\sigma_{11}}{dt} &= ig_s(a\sigma_{e1}^{(j)} - a^*\sigma_{1e}^{(j)}) + \gamma_{\rm th}(\sigma_{ee}^{(j)} + \sigma_{22} + \sigma_{33} - 3\sigma_{11}) + \gamma_{\rm p}\sigma_{ee}^{(j)} \\
\frac{d\sigma_{22}}{dt} &= -i\Omega_2(\sigma_{2e}^{(j)} - \sigma_{e2}^{(j)}) + \gamma_{\rm th}(\sigma_{ee}^{(j)} + \sigma_{11} + \sigma_{33} - 3\sigma_{22}) + \gamma_{\rm p}\sigma_{33} \\
\frac{d\sigma_{33}}{dt} &= \gamma_{\rm th}(\sigma_{11} + \sigma_{22} + \sigma_{ee}^{(j)} - 3\sigma_{33}) - \gamma_{\rm p}\sigma_{33}
\end{aligned} \tag{18}$$

Here $\gamma_{\rm nd}$ is the nuclear spin dephasing rate, $\gamma_{\rm th}$ is the electron spin thermalization rate, Γ is the electron spin dephasing rate, and $\gamma_{\rm p}$ is electron spin polarization rate [3]. In this section, we assume the ensemble is homogeneous ($\Delta_s^{(j)} \to \Delta_s$), while in the next section we will demonstrate this assumption will not affect the solution for the two-field interference.

We now make a semi-classical approximation for the system given that we have a very large number of spins [3]. This involves assuming that the state of the cavity remains close to a coherent state so that correlations between spin operators and the field are negligible. This means that expectation values of products of the cavity mode operator and any spin operator factor simply into products of their respective expectation values. Now, taking expectation values on both sides of the dynamical Eq. (18), all the operators become complex numbers. We will use the same symbols to represent the spin degrees of freedom, even though they are now complex numbers, but we will use α in place of a, which now is the coherent amplitude of the cavity mode. Because $\sigma_{\rm 1e}$ is now a complex number, and because the equations are the same for the $\sigma_{\rm 1e}$ for every spin, we can make the replacement

$$\sum_j g_s\sigma_{\rm 1e}^{(j)} \to Ng_s\sigma_{\rm 1e}. \tag{19}$$



Supplementary Figure 3. **Comparison between analytical and numerical simulations.** We plot the imaginary part of reflection coefficient $\mathrm{Im}(r)$ as a function of detuning Δ with the methods of analytical derivations (blue) and numerical simulations (red). Parameters: $\kappa = 2\pi \times 2.6$ MHz, $\kappa_{\mathrm{c1}} = 2\pi \times 1.3$ MHz, $g_s = 2\pi \times 0.06$ Hz, $N = 2 \times 10^{14}$, $\Gamma = 2\pi \times 330$ kHz, $\gamma_{\mathrm{th}} = 2\pi\times$ 0.08 kHz.

Under the homogeneous and semi-classical assumptions the equations of motion simplify to

$$\begin{aligned}
\frac{d\alpha}{dt} &= -i\Delta\alpha - ig_s N\sigma_{1e} + J - \frac{\kappa}{2}\alpha \\
\frac{d\sigma_{1e}}{dt} &= -i\Delta_s\sigma_{1e} - ig_s\alpha(\sigma_{11} - \sigma_{ee}) - i\Omega_2\sigma_{12} - \frac{\Gamma}{2}\sigma_{1e} \\
\frac{d\sigma_{2e}}{dt} &= -i(\Delta_s - \Delta_2)\sigma_{2e} - ig_s\alpha\sigma_{21} - i\Omega_2(\sigma_{22} - \sigma_{ee}) - \frac{\Gamma}{2}\sigma_{2e} \\
\frac{d\sigma_{12}}{dt} &= -i\Delta_2\sigma_{12} + ig_s\alpha\sigma_{e2} - i\Omega_2\sigma_{1e} - \frac{\gamma_{\mathrm{nd}}}{2}\sigma_{12} \\
\frac{d\sigma_{11}}{dt} &= ig_s(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) + \gamma_{\mathrm{th}}(\sigma_{ee} + \sigma_{22} + \sigma_{33} - 3\sigma_{11}) + \gamma_{\mathrm{p}}\sigma_{ee} \\
\frac{d\sigma_{22}}{dt} &= -i\Omega_2(\sigma_{2e} - \sigma_{e2}) + \gamma_{\mathrm{th}}(\sigma_{ee} + \sigma_{11} + \sigma_{33} - 3\sigma_{22}) + \gamma_{\mathrm{p}}\sigma_{33} \\
\frac{d\sigma_{33}}{dt} &= \gamma_{\mathrm{th}}(\sigma_{11} + \sigma_{22} + \sigma_{ee} - 3\sigma_{33}) - \gamma_{\mathrm{p}}\sigma_{33}
\end{aligned} \tag{20}$$

where we need to remember that all the dynamical variables are now complex numbers not operators. The equations are now for the quantities associated with a single spin, and the only place that the number of spins, N, enters is in the first equation (the equation for the cavity coherent amplitude α).

From the subequations (5-7) and $\sigma_{11} + \sigma_{22} + \sigma_{33} + \sigma_{ee} = 1$, we can get:

$$\begin{aligned}
\sigma_{22} - \sigma_{ee} &= -\frac{i\Omega_2(\sigma_{2e} - \sigma_{e2})}{4\gamma_{\mathrm{th}}} + \frac{ig_s(\alpha\sigma_{e1} - \alpha^*\sigma_{1e})}{\gamma_{\mathrm{p}}} + \frac{1}{2} \\
\sigma_{11} - \sigma_{ee} &= \frac{i\Omega_2(\sigma_{2e} - \sigma_{e2})}{4\gamma_{\mathrm{th}}} + \frac{2ig_s(\alpha\sigma_{e1} - \alpha^*\sigma_{1e})}{\gamma_{\mathrm{p}}} + \frac{1}{2}
\end{aligned} \tag{21}$$

This is with the approximation where $\gamma_p \gg \gamma_{\mathrm{th}}$. We note that this approximation only holds in the EIT regime but not the MWI regime. To get the solution for Eq. (18) in the MWI regime, an ODE solver is used to generate the

figures in the main text. From subequation (4) we get:

$$\sigma_{12} = \frac{i(g_s\alpha\sigma_{e2} - \Omega_2\sigma_{1e})}{\gamma_{\rm nd}/2 - i\Delta_2} \tag{22}$$

From subequation (2-3) we get:

$$\begin{aligned}(\frac{\Gamma}{2} + i\Delta_s + i\Delta + \frac{g_s^2|\alpha|^2}{\gamma_{\rm th}/2 - i\Delta})\sigma_{2e} - \frac{g_s\alpha}{\gamma_{\rm th}/2 - i\Delta}\Omega_2\sigma_{e1} &= -\frac{\Omega_2^2}{4\gamma_{\rm th}}(\sigma_{2e} - \sigma_{e2}) + \frac{\Omega_2 g_s}{\gamma_{\rm p}}(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) - \frac{i\Omega_2}{2} \\ (\frac{\Gamma}{2} + i\Delta_s + \frac{\Omega_2^2}{\gamma_{\rm th}/2 - i\Delta})\sigma_{1e} - \frac{g_s\alpha}{\gamma_{\rm th}/2 - i\Delta}\Omega_2\sigma_{e2} &= \frac{\Omega_2 g_s\alpha}{4\gamma_{\rm th}}(\sigma_{2e} - \sigma_{e2}) + \frac{2g_s^2\alpha}{\gamma_{\rm p}}(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) - \frac{ig_s\alpha}{2}\end{aligned} \tag{23}$$

The conjugate for Eq. (15-1) is:

$$(\frac{\Gamma}{2} - i\Delta_s - i\Delta + \frac{g_s^2|\alpha|^2}{\gamma_{\rm th}/2 + i\Delta})\sigma_{e2} - \frac{g_s\alpha^*}{\gamma_{\rm th}/2 + i\Delta}\Omega_2\sigma_{1e} = \frac{\Omega_2^2}{4\gamma_{\rm th}}(\sigma_{2e} - \sigma_{e2}) - \frac{\Omega_2 g_s}{\gamma_{\rm p}}(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) + \frac{i\Omega_2}{2} \tag{24}$$

Combining Eq. (15-1) and Eq. (16), we can get:

$$\sigma_{e2} = -\frac{f_1}{f_1^*}\sigma_{2e} + \frac{f_2}{f_1^*} \tag{25}$$

with:

$$\begin{aligned} f_1 &= \frac{\Gamma}{2} + i\Delta_s + i\Delta + \frac{g_s^2|\alpha|^2}{\gamma_{\rm th}/2 - i\Delta} \\ f_2 &= \frac{\Omega_2 g_s\alpha\sigma_{e1}}{\gamma_{\rm th}/2 - i\Delta} + \frac{\Omega_2 g_s\alpha^*\sigma_{1e}}{\gamma_{\rm th}/2 + i\Delta}\end{aligned} \tag{26}$$

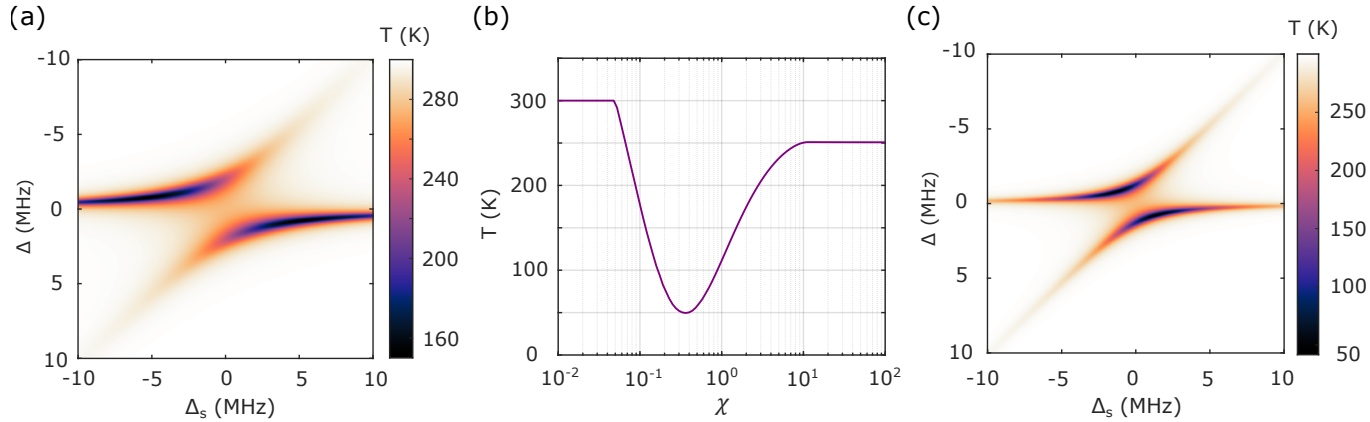
$$\sigma_{2e} = \frac{\frac{\Omega_2^2 f_2}{4\gamma_{\rm th} f_1^*} + \frac{g_s\alpha}{\gamma_{\rm th}/2 - i\Delta}\Omega_2\sigma_{e1} + \frac{\Omega_2 g_s}{\gamma_{\rm p}}(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) - \frac{1}{2}i\Omega_2}{f_1 + \frac{\Omega_2^2}{4\gamma_{\rm th}}(1 + \frac{f_1}{f_1^*})} \tag{27}$$

We plot the reflection $|r|^2$ as a function of Δ within different regimes. First, we set the system in the bad cavity regime where $g < \kappa$ but cooperativity $C > 1$. A dipole-induced reflection (DIR) shows up with the electron spin linewidth around the zero detuning. Second, we set the system in the regime of $g > \max\{\kappa, \Gamma\}$ (strong coupling regime); instead of DIR, two polaritons with avoid-crossing feature show up with splitting $2g$.

We compare the analytical derivations with direct numerical simulation of the time dynamics of Eq. (18). We plot the imaginary part of the reflection coefficient $\mathrm{Im}(r)$ is shown in Fig. S3. The difference between these two methods are within a few percent. The difference between these two methods could be attributed to the approximation we made during the analytical derivations, e.g. we made the approximation that the laser polarization rate ($2\pi \times 1.4$ kHz) is much larger than the thermalization rate ($2\pi \times 0.08$ kHz).

C. Threshold for the spin driving to populate the cavity mode

We find that when the driving of the spin at rate Ω_2 is strong enough, it will feed photons into the cavity, thus providing a non-zero cavity occupation even without a cavity drive. To determine the value of Ω_2 required to do this, we can solve the Maxwell-Bloch equations without the cavity drive and set all detunings to zero. (The absence of a cavity drive simplifies the equations considerably and allows analytic solutions.)



Supplementary Figure 4. **Spin refrigeration.** (a) Cooling temperature T as a function of Δ and Δ_s with experimental parameters [3, 4]: $\kappa = 2\pi \times 0.26$ MHz, $\kappa_{c1} = 2\pi \times 0.13$ MHz, $g_s = 2\pi \times 0.019$ Hz, $N = 2 \times 10^{14}$, $\Gamma = 2\pi \times 330$ kHz. (b) Cooling temperature for different χ. (c) The optimal cooling spectrum with $\chi = 0.36$.

The Maxwell-Bloch equations, without driving, and using that fact that $\sigma_{11} + \sigma_{22} + \sigma_{33} + \sigma_{ee} = 1$, are

$$\begin{aligned}
\frac{d\alpha}{dt} &= -i\Delta\alpha - ig_s N\sigma_{1e} - \frac{\kappa}{2}\alpha \\
\frac{d\sigma_{1e}}{dt} &= -i\Delta_s\sigma_{1e} - ig_s\alpha(\sigma_{11} - \sigma_{ee}) - i\Omega_2\sigma_{12} - \frac{\Gamma}{2}\sigma_{1e} \\
\frac{d\sigma_{2e}}{dt} &= -i(\Delta_s - \Delta_2)\sigma_{2e} - ig_s\alpha\sigma_{21} - i\Omega_2(\sigma_{22} - \sigma_{ee}) - \frac{\Gamma}{2}\sigma_{2e} \\
\frac{d\sigma_{12}}{dt} &= -i\Delta_2\sigma_{12} + ig_s\alpha\sigma_{e2} - i\Omega_2\sigma_{1e} - \frac{\gamma_{\rm nd}}{2}\sigma_{12} \\
\frac{d\sigma_{11}}{dt} &= ig_s(\alpha\sigma_{e1} - \alpha^*\sigma_{1e}) + \gamma_{\rm th}(\sigma_{ee} + \sigma_{22} + \sigma_{33} - 3\sigma_{11}) + \gamma_{\rm p}\sigma_{ee} \\
\frac{d\sigma_{22}}{dt} &= -i\Omega_2(\sigma_{2e} - \sigma_{e2}) + \gamma_{\rm th}(\sigma_{ee} + \sigma_{11} + \sigma_{33} - 3\sigma_{22}) + \gamma_{\rm p}\sigma_{33} \\
\frac{d\sigma_{33}}{dt} &= \gamma_{\rm th}(\sigma_{11} + \sigma_{22} + \sigma_{ee} - 3\sigma_{33}) - \gamma_{\rm p}\sigma_{33}
\end{aligned} \tag{28}$$

Since the final equation only contains σ_{33} we have

$$\sigma_{33} = \frac{\gamma_{\rm th}}{4\gamma_{\rm th} + \gamma_{\rm p}} \equiv r \tag{29}$$

Since the first equation contains only α and σ_{1e} we can write the latter in terms of α. When all the detunings equal zero we have

$$\sigma_{1e} = i\frac{\kappa}{2Ng_s}\alpha \equiv iK\alpha \tag{30}$$

Applying these simplifications, and taking α to be real (any steady-state solution remains valid under the "gauge"

transformation $(\alpha, \sigma_{1e}, J, \sigma_{12}) \to e^{i\theta}(\alpha, \sigma_{1e}, J, \sigma_{12})$)

$$\begin{aligned}
0 &= -ig_s\alpha(r + 2\sigma_{11} + \sigma_{22} - 1) - i\Omega_2\sigma_{12} - i\frac{K\Gamma}{2}\alpha \\
0 &= -ig_s\alpha\sigma_{21} - i\Omega_2(r + \sigma_{11} + 2\sigma_{22} - 1) - \frac{\Gamma}{2}\sigma_{2e} \\
0 &= ig_s\alpha\sigma_{e2} + \Omega_2 K\alpha - \frac{\gamma_{\rm nd}}{2}\sigma_{12} \\
0 &= 2g_s K|\alpha|^2 + \gamma_{\rm th}(1 - 4\sigma_{11}) + \gamma_{\rm p}(1 - r - \sigma_{11} - \sigma_{22}) \\
0 &= -i\Omega_2(\sigma_{2e} - \sigma_{e2}) + \gamma_{\rm th}(1 - 4\sigma_{22}) + r\gamma_{\rm p}
\end{aligned} \tag{31}$$

Using the third equation to eliminate σ_{12} we get:

$$\begin{aligned}
0 &= -ig_s\alpha(r + 2\sigma_{11} + \sigma_{22} - 1) - i\Omega_2\left(i\frac{2g_s}{\gamma_{\rm nd}}\alpha\sigma_{e2} + 2\frac{\Omega_2 K}{\gamma_{\rm nd}}\alpha\right) - i\frac{K\Gamma}{2}\alpha \\
0 &= -ig_s\alpha\left(i\frac{2g_s}{\gamma_{\rm nd}}\alpha\sigma_{e2} + 2\frac{\Omega_2 K}{\gamma_{\rm nd}}\alpha\right)^* - i\Omega_2(r + \sigma_{11} + 2\sigma_{22} - 1) - \frac{\Gamma}{2}\sigma_{2e} \\
0 &= 2g_s K|\alpha|^2 + \gamma_{\rm th}(1 - 4\sigma_{11}) + \gamma_{\rm p}(1 - r - \sigma_{11} - \sigma_{22}) \\
0 &= -i\Omega_2(\sigma_{2e} - \sigma_{e2}) + \gamma_{\rm th}(1 - 4\sigma_{22}) + r\gamma_{\rm p}
\end{aligned} \tag{32}$$

Now we use the first equation to eliminate σ_{2e}:

$$\sigma_{2e} = -i\frac{\gamma_{\rm nd}}{2\Omega_2}(r + 2\sigma_{11} + \sigma_{22} - 1) - i\frac{\gamma_{\rm nd}}{2g_s\Omega_2}\left(\frac{2K\Omega_2^2}{\gamma_{\rm th}} + \frac{K\Gamma}{2}\right) \equiv -iA(r + 2\sigma_{11} + \sigma_{22} - 1) - iB \tag{33}$$

Eliminating σ_{2e} we have

$$\begin{aligned}
0 &= -\frac{2g_s^2}{\gamma_{\rm nd}}|\alpha|^2[-iA(r + 2\sigma_{11} + \sigma_{22} - 1) - iB] - i\frac{2g_s\Omega_2 K}{\gamma_{\rm nd}}|\alpha|^2 - i\Omega_2(r + \sigma_{11} + 2\sigma_{22} - 1) - \frac{\Gamma}{2}[-iA(r + 2\sigma_{11} + \sigma_{22} - 1) - iB] \\
0 &= 2g_s K|\alpha|^2 + \gamma_{\rm th}(1 - 4\sigma_{11}) + \gamma_{\rm p}(1 - r - \sigma_{11} - \sigma_{22}) \\
0 &= -2\Omega_2 A(r + 2\sigma_{11} + \sigma_{22} - 1) - 2\Omega_2 B + \gamma_{\rm th}(1 - 4\sigma_{22}) + r\gamma_{\rm p}
\end{aligned} \tag{34}$$

Simplifying:

$$\begin{aligned}
0 &= \frac{2g_s^2}{\gamma_{\rm nd}}|\alpha|^2[A(r + 2\sigma_{11} + \sigma_{22} - 1) + B] - \frac{2g_s\Omega_2 K}{\gamma_{\rm nd}}|\alpha|^2 - \Omega_2(r + \sigma_{11} + 2\sigma_{22} - 1) + \frac{\Gamma}{2}[A(r + 2\sigma_{11} + \sigma_{22} - 1) + B] \\
0 &= -(4\gamma_{\rm th} + \gamma_{\rm p})\sigma_{11} - \gamma_{\rm p}\sigma_{22} + \gamma_{\rm th} + \gamma_{\rm p}(1 - r) + 2g_s K|\alpha|^2 \\
0 &= -4\Omega_2 A\sigma_{11} - (2\Omega_2 A + 4\gamma_{\rm th})\sigma_{22} - 2\Omega_2 B + \gamma_{\rm th} + r\gamma_{\rm p} - 2\Omega_2 A(r - 1)
\end{aligned} \tag{35}$$

finally we have to eliminate σ_{11} and σ_{22}, for which we will use the second and third equations:

$$\begin{aligned}
\sigma_{11} &= -\frac{2\Omega_2 A + 4\gamma_{\rm th}}{4\Omega_2 A}\sigma_{22} + \frac{-2\Omega_2 B + \gamma_{\rm th} + r\gamma_{\rm p} - 2\Omega_2 A(r - 1)}{4\Omega_2 A} \\
&= -\left(\frac{1}{2} + \frac{\gamma_{\rm th}}{\Omega_2 A}\right)\sigma_{22} + \frac{\gamma_{\rm th} + r\gamma_{\rm p}}{4\Omega_2 A} - \frac{(r - 1)}{2} - \frac{B}{2A} \\
&\equiv -\left(\frac{1}{2} + \frac{\gamma_{\rm th}}{\Omega_2 A}\right)\sigma_{22} + C
\end{aligned} \tag{36}$$

$$
\begin{aligned}
0 &= \left[(4\gamma_{\rm th}+\gamma_{\rm p})\left(\frac{1}{2}+\frac{\gamma_{\rm th}}{\Omega_2 A}\right)-\gamma_{\rm p}\right]\sigma_{22}-(4\gamma_{\rm th}+\gamma_{\rm p})C+\gamma_{\rm th}+\gamma_{\rm p}(1-r)+2g_sK|\alpha|^2 \\
\sigma_{22} &= \frac{\gamma_{\rm th}+\gamma_{\rm p}(1-r)-(4\gamma_{\rm th}+\gamma_{\rm p})C}{\left[(4\gamma_{\rm th}+\gamma_{\rm p})\left(\frac{1}{2}+\frac{\gamma_{\rm th}}{\Omega_2 A}\right)-\gamma_{\rm p}\right]}+\frac{2g_sK}{\left[(4\gamma_{\rm th}+\gamma_{\rm p})\left(\frac{1}{2}+\frac{\gamma_{\rm th}}{\Omega_2 A}\right)-\gamma_{\rm p}\right]}|\alpha|^2 \\
&= \frac{\gamma_{\rm th}+\gamma_{\rm p}(1-r)-(4\gamma_{\rm th}+\gamma_{\rm p})C}{10\gamma_{\rm th}+(3/2)\gamma_{\rm p}}+\frac{2g_sK}{10\gamma_{\rm th}+(3/2)\gamma_{\rm p}}|\alpha|^2 \\
&\equiv D+E|\alpha|^2 \\
\sigma_{11} &= C-\left(\frac{1}{2}+\frac{\gamma_{\rm th}}{\Omega_2 A}\right)D-\left(\frac{1}{2}+\frac{\gamma_{\rm th}}{\Omega_2 A}\right)E|\alpha|^2 \equiv C-FD-FE|\alpha|^2
\end{aligned}
\tag{37}
$$

So finally we have to substitute σ_{11} and σ_{22} into the first equation:

$$
0=\frac{2g_s^2}{\gamma_{\rm nd}}|\alpha|^2[A(r+2\sigma_{11}+\sigma_{22}-1)+B]-\frac{2g_s\Omega_2K}{\gamma_{\rm nd}}|\alpha|^2-\Omega_2(r+\sigma_{11}+2\sigma_{22}-1)+\frac{\Gamma}{2}[A(r+2\sigma_{11}+\sigma_{22}-1)+B]
\tag{38}
$$

and we see that we will have a quadratic equation for $|\alpha|^2$. We have

$$
\begin{aligned}
\sigma_{11}+2\sigma_{22} &= C-FD-FE|\alpha|^2+2D+2E|\alpha|^2 = C+(2-F)D+(2-F)\,E|\alpha|^2 \\
2\sigma_{11}+\sigma_{22} &= 2C-2FD-2FE|\alpha|^2+D+E|\alpha|^2 = 2C+(1-2F)D+(1-2F)E|\alpha|^2
\end{aligned}
\tag{39}
$$

So

$$
\begin{aligned}
0 = &\frac{2g_s^2}{\gamma_{\rm nd}}A\left[2C+(1-2F)D+(1-2F)E|\alpha|^2\right]|\alpha|^2+\frac{2g_s^2}{\gamma_{\rm nd}}[A(r-1)+B]|\alpha|^2-\frac{2g_s\Omega_2K}{\gamma_{\rm nd}}|\alpha|^2 \\
&+\Omega_2(1-r)+\frac{\Gamma}{2}[A(r-1)+B] \\
&-\Omega_2\left[C+(2-F)D+(2-F)\,E|\alpha|^2\right]+\frac{A\Gamma}{2}[2C+(1-2F)D+(1-2F)E|\alpha|^2]
\end{aligned}
\tag{40}
$$

Co-efficients of the quadratic:

$$
\begin{aligned}
|\alpha|^4:\ & \frac{2g_s^2}{\gamma_{\rm nd}}A(1-2F)E=-\frac{1}{N}\left(\frac{4g_s^2}{\Omega_2}\right)\left(\frac{\kappa}{10\gamma_{\rm th}+(3/2)\gamma_{\rm p}}\right) \\
&=-\frac{1}{N}\left(\frac{4g_s^2}{\Omega_2}\right)\left(\frac{\kappa}{2\gamma_{\rm p}}\right) \\
|\alpha|^2:\ & \frac{2g_s^2}{\gamma_{\rm nd}}A\left[2C+(1-2F)D\right]+\frac{2g_s^2}{\gamma_{\rm nd}}[A(r-1)+B]-\frac{2g_s\Omega_2K}{\gamma_{\rm nd}}-\Omega_2\left(2-F\right)E+\frac{A\Gamma}{2}(1-2F)E \\
&=\frac{g_s^2}{\Omega_2}\left[2C-4D\right]-\frac{g_s^2}{\Omega_2}\left(\frac{3\gamma_{\rm th}+\gamma_{\rm p}}{4\gamma_{\rm th}+\gamma_{\rm p}}\right)+\frac{\kappa}{N}\left[\frac{\Gamma}{4\Omega_2}+\left(\frac{\Omega_2}{2\gamma_{\rm th}}-\frac{\Gamma}{\Omega_2}\right)\frac{\gamma_{\rm th}}{10\gamma_{\rm th}+(3/2)\gamma_{\rm p}}\right] \\
&=\frac{g_s^2}{\Omega_2}\frac{10}{3}-\frac{\kappa}{N}\left[\frac{14\Omega_2}{3\gamma_{\rm th}}+\frac{\Omega_2}{3\gamma_{\rm p}}+\frac{10\Gamma}{12\Omega_2}\right] \\
1:\ & \Omega_2(1-r)+\frac{\Gamma}{2}[A(r-1)+B]-\Omega_2\left[C+(2-F)D\right]+\frac{A\Gamma}{2}[2C+(1-2F)D] \\
&=-\frac{2}{3}\Omega_2+\frac{4\Gamma\gamma_{\rm th}}{3\Omega_2}-\frac{3\kappa\Gamma\Omega_2}{4Ng_s^2}-\frac{\gamma_{\rm th}\kappa\Gamma^2}{12\Omega_2Ng_s^2}+\frac{8\kappa\Omega_2^3}{3\gamma_{\rm th}Ng_s^2}
\end{aligned}
\tag{41}
$$

and

$$
\begin{aligned}
A &= \frac{\gamma_{\rm nd}}{2\Omega_2} \\
B &= \frac{\kappa\Omega_2}{2Ng_s^2} + \frac{\gamma_{\rm nd}\kappa\Gamma}{8\Omega_2 N g_s^2} \\
C &= \frac{\gamma_{\rm th} + r\gamma_{\rm p}}{4\Omega_2 A} - \frac{(r-1)}{2} - \frac{B}{2A} = \frac{1}{2}\left(\frac{7\gamma_{\rm th} + 3\gamma_{\rm p}}{4\gamma_{\rm th} + \gamma_{\rm p}}\right) - \frac{\kappa}{Ng_s^2}\left[\frac{2\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right] \\
D &= \frac{\gamma_{\rm th}}{10\gamma_{\rm th} + (3/2)\gamma_{\rm p}} + \frac{\gamma_{\rm p}}{10\gamma_{\rm th} + (3/2)\gamma_{\rm p}}\left(\frac{3\gamma_{\rm th} + \gamma_{\rm p}}{4\gamma_{\rm th} + \gamma_{\rm p}}\right) - \frac{4\gamma_{\rm th} + \gamma_{\rm p}}{10\gamma_{\rm th} + (3/2)\gamma_{\rm p}} C \\
&= \frac{4\gamma_{\rm th}^2 + 4\gamma_{\rm p}\gamma_{\rm th} + \gamma_{\rm p}^2}{(10\gamma_{\rm th} + (3/2)\gamma_{\rm p})(4\gamma_{\rm th} + \gamma_{\rm p})} - \frac{7\gamma_{\rm th} + 3\gamma_{\rm p}}{20\gamma_{\rm th} + 3\gamma_{\rm p}} + \frac{4\gamma_{\rm th} + \gamma_{\rm p}}{10\gamma_{\rm th} + (3/2)\gamma_{\rm p}}\frac{\kappa}{Ng_s^2}\left[\frac{2\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right] \\
E &= \frac{\kappa/N}{10\gamma_{\rm th} + (3/2)\gamma_{\rm p}} \\
F &= 5/2 \\
K &= \frac{\kappa}{2Ng_s} \\
r &= \frac{\gamma_{\rm th}}{4\gamma_{\rm th} + \gamma_{\rm p}}
\end{aligned}
\tag{42}
$$

We can simplify the form of these equations by using the fact that some terms are negligible compared to others due to the values of the parameters we use for our system. First, since we have $\gamma_{\rm th} \sim \gamma_{\rm p}/20$ we first simplify the expressions above by using the approximation $\gamma_{\rm th} \ll \gamma_{\rm p}$. This will give results accurate to about $\pm 10\%$. Now C, D, E, and $1-r$ become:

$$
\begin{aligned}
C &= \frac{3}{2} - \frac{\kappa}{Ng_s^2}\left[\frac{2\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right] \\
D &= -1/3 + \frac{2\kappa}{3Ng_s^2}\left[\frac{2\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right] \\
E &= \frac{2\kappa}{3N\gamma_{\rm p}} \\
1 - r &= 1 \\
2C - 4D &= \frac{13}{3} - \frac{14\kappa}{3Ng_s^2}\left[\frac{\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right] \\
C - D/2 &= \frac{5}{3} - \frac{4\kappa}{3Ng_s^2}\left[\frac{2\Omega_2^2}{\gamma_{\rm th}} + \frac{\Gamma}{8}\right]
\end{aligned}
\tag{43}
$$

Now we will write the coefficients in terms of $\tilde{n} = |\alpha|^2/N$. Thus $\tilde{n}$ gives the number of cavity photons as a fraction of the number of NV centers in the cavity.

$$
\begin{aligned}
\tilde{n}^2 &: -\left(\frac{4g^2}{\Omega_2}\right)\left(\frac{\kappa}{2\gamma_{\rm p}}\right) \\
\tilde{n} &: \frac{g^2}{\Omega_2}\frac{10}{3} - \kappa\left[\frac{14\Omega_2}{3\gamma_{\rm th}} + \frac{\Omega_2}{3\gamma_{\rm p}} + \frac{10\Gamma}{12\Omega_2}\right] \\
1 &: -\frac{2}{3}\Omega_2 + \frac{4\Gamma\gamma_{\rm th}}{3\Omega_2} - \frac{3\kappa\Gamma\Omega_2}{4g^2} - \frac{\gamma_{\rm th}\kappa\Gamma^2}{12\Omega_2 g^2} + \frac{8\kappa\Omega_2^3}{3\gamma_{\rm th}g^2}
\end{aligned}
\tag{44}
$$

Now normalizing the coefficients by the coefficient of $\tilde{n}^2$:

$$\begin{aligned}
\tilde{n}^2 :& 1 \\
\tilde{n} :& \frac{\Omega_2^2}{6g^2}\left[1+14\frac{\gamma_\mathrm{p}}{\gamma_\mathrm{th}}\right]-\frac{5\gamma_\mathrm{p}}{6\kappa}\left[1-\frac{1}{2}\frac{\Gamma\kappa}{g^2}\right] \\
1 :& \frac{\gamma_\mathrm{p}\Gamma}{g^2}\left[\frac{1}{3}\frac{\Omega_2^2}{\kappa\Gamma}-\frac{2}{3}\frac{\gamma_\mathrm{th}}{\kappa}+\frac{3}{8}\frac{\Omega_2^2}{g^2}\left[1-\frac{32}{9}\frac{\Omega_2^2}{\gamma_\mathrm{th}\Gamma}\right]+\frac{1}{24}\frac{\gamma_\mathrm{th}\Gamma}{g^2}\right]
\end{aligned} \tag{45}$$

These expressions are written in terms of cooperativities or ratios of coupling or decay rates.

The solution to the quadratic equation for $\tilde{n}$ is

$$\tilde{n} = -\frac{b}{2}+\sqrt{\left(\frac{b}{2}\right)^2-c} \tag{46}$$

with

$$b = \frac{\Omega_2^2}{6g^2}\left[1+14\frac{\gamma_\mathrm{p}}{\gamma_\mathrm{th}}\right]-\frac{5\gamma_\mathrm{p}}{6\kappa}\left[1-\frac{1}{2}\frac{\Gamma\kappa}{g^2}\right] \approx \frac{7}{3}\frac{\Omega_2^2}{g^2}\frac{\gamma_\mathrm{p}}{\gamma_\mathrm{th}}-\frac{5\gamma_\mathrm{p}}{6\kappa}\left[1-\frac{1}{2}\frac{\Gamma\kappa}{g^2}\right] \tag{47}$$

$$= \frac{\gamma_\mathrm{p}}{\gamma_\mathrm{th}}\left[\frac{7}{3}\frac{\Omega_2^2}{g^2}-\frac{5\gamma_\mathrm{th}}{6\kappa}\left[1-\frac{1}{2}\frac{\Gamma\kappa}{g^2}\right]\right] \tag{48}$$

$$c = \frac{\gamma_\mathrm{p}\Gamma}{g^2}\left[\frac{1}{3}\frac{\Omega_2^2}{\kappa\Gamma}-\frac{2}{3}\frac{\gamma_\mathrm{th}}{\kappa}+\frac{3}{8}\frac{\Omega_2^2}{g^2}\left[1-\frac{32}{9}\frac{\Omega_2^2}{\gamma_\mathrm{th}\Gamma}\right]+\frac{1}{24}\frac{\gamma_\mathrm{th}\Gamma}{g^2}\right] \tag{49}$$

$$= -\frac{4}{3}\frac{\gamma_\mathrm{p}}{\gamma_\mathrm{th}}\left(\frac{\Omega_2^2}{g^2}\right)^2+\frac{\gamma_\mathrm{p}\Gamma}{g^2}\left[\frac{1}{3}\frac{\Omega_2^2}{\kappa\Gamma}-\frac{2}{3}\frac{\gamma_\mathrm{th}}{\kappa}+\frac{3}{8}\frac{\Omega_2^2}{g^2}+\frac{1}{24}\frac{\gamma_\mathrm{th}\Gamma}{g^2}\right] \tag{50}$$

So to have a real solution we must have

$$\left(\frac{b}{2}\right)^2 \geq c \tag{51}$$

and to have a non-zero cavity occupation we must have

$$\left(\frac{b}{2}\right)^2 - c > \frac{b}{2} \tag{52}$$

So long as there is a real solution, we will always have a positive cavity occupation if $b < 0$ which means

$$\frac{\gamma_\mathrm{th}}{\kappa}\left[1-\frac{1}{2}\frac{\Gamma\kappa}{g^2}\right] > \frac{42}{15}\frac{\Omega_2^2}{g^2} \tag{53}$$

But in our system b is positive so we have to satisfy (52).

We need the actual values of the parameters to tell what terms are important:

$$g^2 = Ng_\mathrm{s}^2 = 1.2\times10^{15}\times(2\pi\times0.06\ \mathrm{Hz})^2 = (2\pi\times2.08\ \mathrm{MHz})^2 \tag{54}$$

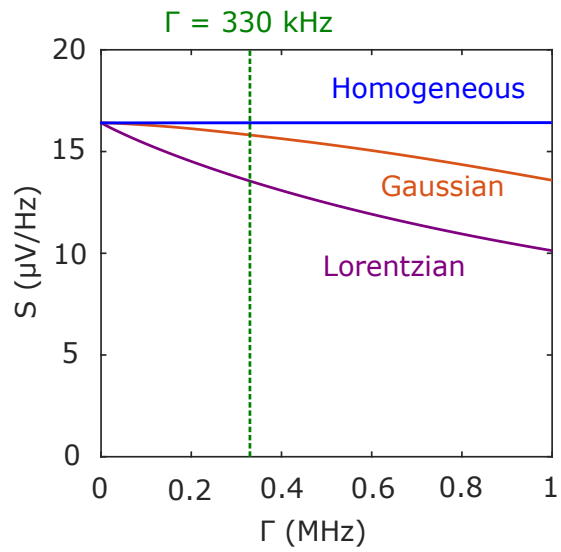
$$\kappa = 2\pi\times2.6\ \mathrm{MHz} \tag{55}$$

$$\Gamma = 2\pi\times0.33\ \mathrm{MHz} \tag{56}$$

$$\gamma_\mathrm{p} = 2\pi\times1.4\times10^{-3}\ \mathrm{MHz} \tag{57}$$

$$\gamma_\mathrm{th} = 2\pi\times80\ \mathrm{Hz} \tag{58}$$

$$\gamma_\mathrm{nd} = 2\pi\times80\ \mathrm{Hz} \tag{59}$$



Supplementary Figure 5. **Comparison of different inhomogeneouty assumptions.** We plot the signal S as a function of inhomogeneous linewidth, assuming the ensemble follows either a Gaussian (red) or Lorentzian (purple) distribution. We also plot the solution assuming homogeneous ensemble as a comparison (blue). At an inhomogeneous linewidth $\Gamma = 330$ kHz, the signal changes by only 4% under the assumption of a Gaussian distribution.

So if we assume that the threshold region will be when Ω_2 is not very much smaller than g, then the only terms that contribute appreciably are

$$b = \frac{\gamma_{\rm p}}{\gamma_{\rm th}} \frac{7}{3} \frac{\Omega_2^2}{g^2} \tag{60}$$

$$c = -\frac{4}{3} \frac{\gamma_{\rm p}}{\gamma_{\rm th}} \left(\frac{\Omega_2^2}{g^2} \right)^2 \tag{61}$$

So the condition for non-zero cavity amplitude is

$$\left[\frac{\gamma_{\rm p}}{\gamma_{\rm th}} \left(\frac{7}{6} \right)^2 + \frac{4}{3} \right] \frac{\Omega_2^2}{g^2} \approx \frac{4}{3} \left[\frac{\gamma_{\rm p}}{\gamma_{\rm th}} + 1 \right] \frac{\Omega_2^2}{g^2} > \frac{7}{6} \tag{62}$$

or

$$\Omega_2 > g \sqrt{\frac{7/8}{\left[\frac{\gamma_{\rm p}}{\gamma_{\rm th}} + 1 \right]}} \approx \sqrt{\frac{\gamma_{\rm th}}{\gamma_{\rm p}}} g \approx \frac{1}{4} g \tag{63}$$

So, given the parameter regime of our system, the threshold for Ω_2 as a fraction of the coupling g only depends on $\gamma_{\rm th}$ and $\gamma_{\rm p}$.

D. Discussion of the effects of inhomogeneity of the spins

In the electron spin-based NV-cQED system, inhomogeneity plays a central role in determining the signal response and sensitivity [3]. However, in our nuclear spin-based system, we have assumed a homogeneous linewidth in our analysis, in particular using the dynamical Eq. (20). We now provide numerical evidence that this is a reasonable assumption, meaning that inhomogeneity at the level typically seen in NV ensembles will not change the signal by more than a few percent.

We consider two kinds of inhomogeneous distributions for the frequencies of the ensemble members, Gaussian and Lorentzian. The first is expected to be the most realistic assumption, and we consider the other as a rather extreme example of a distribution that has long tails. The Gaussian distribution is

$$P_{\mathrm{G}}(\omega) = \frac{1}{\sqrt{\pi}\Gamma_1/\eta} e^{-\left(\frac{\omega-\omega_{\mathrm{s}}}{\Gamma_1/\eta}\right)^2}, \tag{64}$$

and the Lorentzian is

$$P_{\mathrm{L}}(\omega) = \frac{1}{2\pi\Gamma_1} \frac{\Gamma_1^2}{(\omega-\omega_{\mathrm{s}})^2 + \Gamma_1^2/4}, \tag{65}$$

with ω_s the center frequency of the ensemble distribution, Γ_1 the inhomogeneous linewidth, and the factor $\eta = 2\sqrt{\ln 2}$ relates the Gaussian standard deviation to a Lorentzian full-width half-max.

It is challenging numerically to solve Eq. (17) rigorously since it contains 3×10^{14} equations for a spin ensemble of $N = 10^{14}$. So we use an approximate procedure to estimate the effects of inhomogeneity. First we take 201 values for the spin detunings equispaced over the range covered by the distributions, and solve the homogeneous equations (Eqs. (28)) in the semiclassical limit separately for each value of the detuning. We find that the solution of the cavity photon number is almost the same for the detunings which are much smaller than the cavity linewidth. The homogeneous ensemble approximation is then valid when (1) the inhomogeneous linewidth of the electron spin is much smaller than the cavity bandwidth κ, and (2) the system is in the linear regime where the spins remain close to their ground states. This is the case for the results in the main text.

Note that to solve the inhomogeneous equations more exactly, the values of $\langle\sigma_{1\mathrm{e}}^{(j)}\rangle$ for each value of the detuning should be averaged to produce the term $\sum_j g_s\sigma_{1\mathrm{e}}^{(j)}$ in the equation for α. So as a self-consistency check, we calculate this average over the 201 values of the detuning. The result differs from the value for the homogeneous situation by only about 1%.

Averaging the results for each of the detunings over each of the two distributions produces the signals plotted in Fig. S5. We observe that the signal differs depending on the assumed type of inhomogeneity. Notably NV ensembles are more accurately modeled as Gaussian-distributed [5], which results in only a 4% deviation from the homogeneous ensemble at the experimentally relevant value of $\Gamma_1 = 330$ kHz (green dash line on the plot). This indicates that our homogeneous treatment is sufficiently accurate to demonstrate the efficacy of our EIT metrology scheme.

One might intuitively think that the inhomogeneous distribution plays a central role in the nuclear spin gyroscope, as highlighted in our previous NV-cQED work on electron spin readout [3]. However, in the two-field interference regime, it is the frequency difference between the driving field and the cavity probe field that matters, rather than the absolute frequency. EIT will still occur in the presence of common detuning, as long as the condition $\omega_2 + |Q| - \omega_d$ is satisfied. This remains independent of the spin frequency when the inhomogeneous broadening is much smaller than the cavity bandwidth, as is the case in this work.

E. Spin refrigeration

In the main text, we assume a 2 dB spin refrigeration rate for our model, see Fig. S4(a). Here, we will give more details about the tradeoff on the mode cooling on the parameter space. We use a perturbation model for the

steady-state cooling. The linear theory gives the cavity output power spectrum as [4]:

$$N_P(\omega) = n_T + (n_\mathrm{c} - n_T)\, C, \tag{66}$$

where $n_T = [\exp(\hbar\omega/k_B T) - 1]^{-1}$ is the room temperature photon number and the effective "cold spin bath" photon number n_c. The definition of C is:

$$C = \frac{\kappa_\mathrm{c}\Gamma g_\mathrm{eff}^2}{\left(\frac{\Gamma^2}{4} + \Delta^2\right)\left[\frac{\kappa^2}{4} + \Delta_\mathrm{s}^2\right] + g_\mathrm{eff}^2[\Gamma\frac{\kappa}{2} - 2\Delta\Delta_\mathrm{s}] + g_\mathrm{eff}^4}. \tag{67}$$

We plot the temperature $T \sim (1-C)n_T$ as a function of Δ and Δ_s for the parameter set: $\kappa = 2\pi \times 2.6$ MHz, $\kappa_{\mathrm{c}1} = 2\pi \times 1.3$ MHz, $g_s = 2\pi \times 0.06$ Hz, $N = 1.2 \times 10^{15}$, $\Gamma = 2\pi \times 330$ kHz. We observe that the mode cooling happens at the polariton with a maximum cooling of $\Delta T = 145$ K. Now we keep the cooperativity to be the same and set a tuning factor χ to let $\kappa' = \chi\kappa$ and $g_s' = \sqrt{\chi} g_s$. For simplicity, we assume $\kappa_{c0} = 130$ kHz based on the experiment parameter. We plot the minimum temperature with the requirement $\Delta = \Delta_s$ (EIT condition) in Fig. S4(b). A minimum temperature $T = 50$ K at $\chi = 0.36$. We plot the spectrum for the optimal parameter setting in Fig. S4(c).

F. Crosstalk in comagnetometry and vector gyroscope

We present the sensitivity after considering the crosstalk induced by densely packed probes and driving fields. The crosstalk is mainly induced by the Rabi driving for the EIT driving field or the comagnetometer probe field, etc. We assume the crosstalk frequency has a detuning Δ_T with the EIT probe field and a Rabi driving frequency Ω_R. Assuming the Rabi frequency for the EIT probe field is Ω_2, the equivalent Rabi driving frequency with this Raman-type transition can be expressed as:

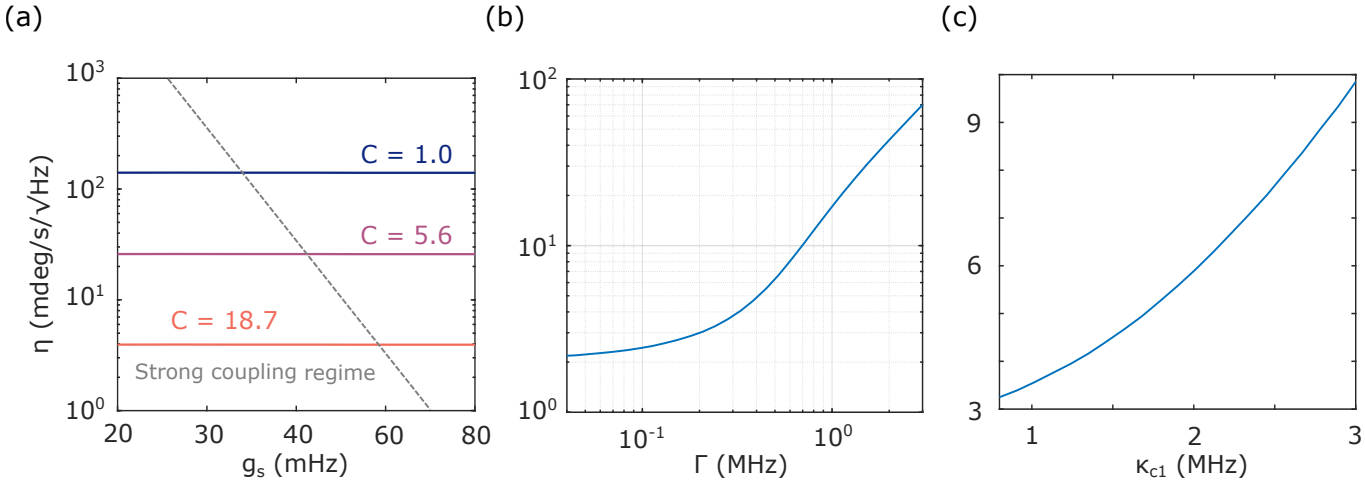
$$\Omega_\mathrm{eq} = \frac{\Omega_R \Omega}{\Delta_T} \tag{68}$$

This equivalent Ω_eq will depolarize the NV preserving transition and degrade the performance, as shown in Fig. 3b in the main text. The equivalent Rabi rotation adds to the Rabi frequency generated by the cavity probe $g_s\alpha$, which deviates the system from the ideal condition.

III. SENSITIVITY OPTIMIZATION

In the main text, we mainly discuss the case with $C \sim 20$. The parameters we used for the discussion is: $\kappa = 2\pi \times 2.6$ MHz, $\kappa_{\mathrm{c}1} = 2\pi \times 1.3$ MHz, $g_s = 2\pi \times 0.06$ Hz, $N = 1.2 \times 10^{15}$, $\Gamma = 2\pi \times 330$ kHz, and nuclear spin dephasing time $T_2^* = 2$ ms (or corresponding rate $\Gamma_n = 2\pi \times 80$ Hz). We plot the sensitivity dependence on the cooperativity in the main text. In this section, we provide more sensitivity discussions related to other parameters.

The sensitivity is independent of the Q'/V for different cavity designs with Q' the loaded quality factor and V the mode volume. In Fig. S6(a), we fix the cooperativity at $C = 1.0$, 5.6, and 18.7 while changing the coupling strength $g_s \to \sqrt{\chi} g_s$ and cavity linewidth $\kappa \to \chi\kappa$ with χ a tuning factor. The sensitivity is independent of the g_s^2/κ for a fixed cooperativity. We then plot the sensitivity dependence with other spin-cavity parameters κ and Γ in Fig. S6(b),(c). The sensitivity scale linearly with the inhomogeneous linewidth of the NV ensemble within $\Gamma > 1$ MHz but becoming



Supplementary Figure 6. **Sensitivity dependence on spin-cavity parameters.** (a) Sensitivity as a function of single spin coupling strength g_s give fixed cooperativites $C = 1.0$, $C = 5.6$, and $C = 18.7$. The strong coupling regime is marked with a dashed line, and the sensitivity is independent of whether the system is in the strong coupling regime or the bad cavity regime. (b) Sensitivity as a function of inhomogeneous linewidth Γ. (c) Sensitivity as a function of cavity coupling strength κ_{c1} assuming critical coupling condition $\kappa_c = \kappa_{c1}$. Parameters in this plot: $\kappa = 2\pi \times 2.6$ MHz, $\kappa_{c1} = 2\pi \times 1.3$ MHz, $g_s = 2\pi \times 0.06$ Hz, $N = 1.2 \times 10^{15}$, $\Gamma = 2\pi \times 330$ kHz, and nuclear spin dephasing time $T_2^* = 2$ ms (or corresponding rate $\Gamma_n = 2\pi \times 80$ Hz), $\Omega_2 = 6$ kHz. $P = -55$ dBm.

saturated for small inhomogeneous linewidth. This is because when the inhomogeneous linewidth is small enough, $t(\Delta = 0)$ will be approaching zero when the driving field is absent; further improving the inhomogeneous linewidth will not then lead to an enhanced contrast for the EIT or MWI. The sensitivity dependence for κ_{c1} is similar to the inhomogeneous linewidth, as shown in Fig. S6(c). The only difference is that decreasing κ_{c1} will lead the system to the oscillation regime, which is undesirable for the sensing technique.

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